

Coplots of a Fit to the Rubber Data and Cropping

There are only 30 observations of the rubber-specimen variables and three of them are aberrant. Because there is so little data, we will begin the fitting with a parametric family that is as simple as is reasonable. And we will use bisquare fitting in the hope that it will prevent the aberrant observations from distorting the fit to the remaining.

The coplots of the rubber data in Figures 4.3 and 4.4 showed a dependence on tensile strength that has a hockey-stick pattern with a break at 180 kg/cm^2 , and a dependence on hardness that is linear. Furthermore, there appeared to be little or no interaction between the factors. We need a mathematical description of this dependence. Let t denote tensile strength. Let

$$[t - 180]^- = \begin{cases} t - 180 & \text{for } t < 180 \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, let

$$[t - 180]^+ = \begin{cases} t - 180 & \text{for } t > 180 \\ 0 & \text{otherwise.} \end{cases}$$

The hockey-stick pattern is

$$\gamma[t - 180]^- + \delta[t - 180]^+.$$

γ is the slope for $t < 180$ and δ is the slope for $t > 180$. Let h denote hardness. The observed dependence of abrasion loss on hardness and tensile strength is described by the function

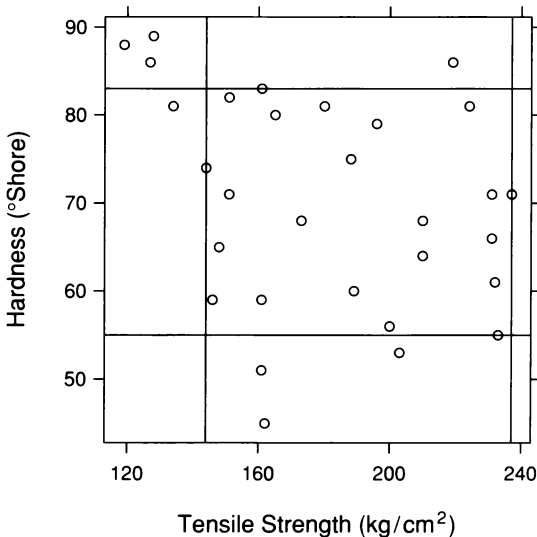
$$g(h, t) = \mu + \beta h + \gamma[t - 180]^- + \delta[t - 180]^+.$$

We will begin by forcing δ to be zero since it appeared to be zero on the coplots. Thus the parametric family has three free parameters: μ , β , and γ . The bisquare fit of the family is

$$\hat{g}(h, t) = 602 - 6.74h - 3.37[t - 180]^-.$$

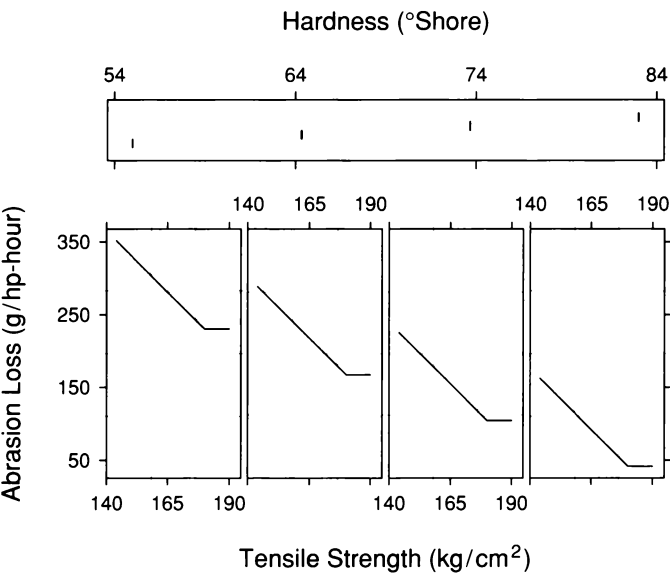
Great care must be exercised in selecting the region of the factor space over which we study a fitted surface. For the ethanol data, the fit was visualized over the smallest box that contains the observations of the two factors, the data rectangle. This was reasonable because most of the rectangle is populated with observations. The rubber data are quite a different case. Figure 4.15 graphs the observations of hardness and tensile strength. The data rectangle has regions not populated with data. Visualizing the fit over the data rectangle would show behavior not supported by the data.

Figure 4.15 has two pairs of parallel lines. We will study the fit only over the intersection of the two regions between the two pairs of lines. This *cropping* of the data rectangle, which will be discussed in detail in Chapter 5, was carried out visually to produce a region well populated throughout with data. The cropped region does not include the upper left corner of the data rectangle. Because the data there are isolated and aberrant, we cannot expect to determine the dependence of the response on the factors in that corner with much precision. Note that the cropping does not imply that data are discarded outside of the cropped region; we still use all of the data in the bisquare fitting.

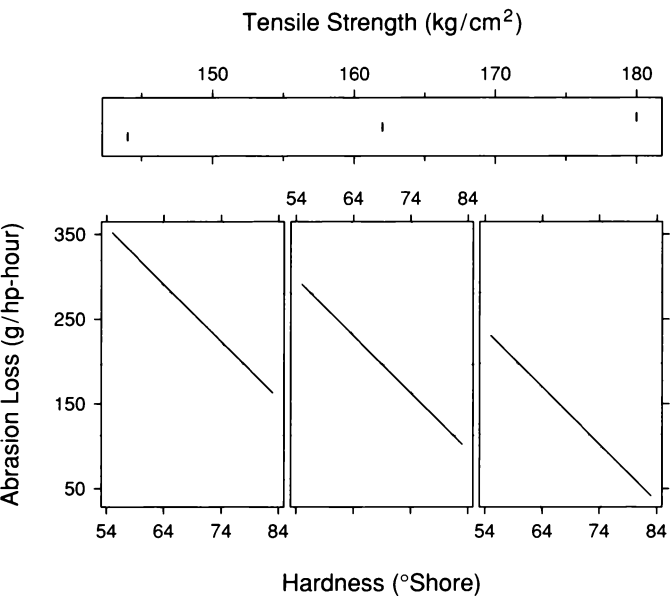


4.15 The rectangle formed by the intersection of the region between the two vertical lines and the region between the two horizontal lines is the cropped region for evaluating the bisquare fit to the rubber data.

Figures 4.16 and 4.17 are coplots of the fit for values of hardness and tensile strength within the cropped region. In Figure 4.16, the four conditioning values of hardness span the full range of the cropped region, but the values of tensile strength are cut off at 190 kg/cm^2 because the fit on each panel is constant above 180 kg/cm^2 . In Figure 4.17, the largest conditioning value of tensile strength is 180 kg/cm^2 , because for any larger value the fit as a function of hardness is the same as that for tensile strength equal to 180 kg/cm^2 . The four hockey-stick curves in Figure 4.16 are parallel and move up or down as we move from one dependence panel to another. This is also true of the three lines in Figure 4.17. These parallel patterns occur, of course, because the parametric function fitted to the data contains no term that allows for an interaction. The residuals for this fit will be visualized in Section 4.4 to determine whether the fit accurately reflects patterns in the data.



4.16 A coplot graphs the fitted abrasion loss surface against tensile strength given hardness. The segments of the fits below 180 kg/cm², the breakpoint, are banked to 45°.



4.17 A coplot graphs the fitted abrasion loss surface against hardness given tensile strength. The lines are banked to 45°.

For the Record: The Details of Loess Fitting

Let the measurements of the two factors be denoted by $x_i = (u_i, v_i)$. If standardization is carried out as described above, then the x_i are the standardized values. Let $x = (u, v)$ be a point in the space of the factors at which the loess surface is to be computed. Again, if the measurements of the factors are standardized, x lies in the standardized space of the factors.

Suppose first that $\alpha \leq 1$. As with one factor, q is αn truncated to an integer, and the neighborhood weight given to the observation (x_i, y_i) for the fit at x is

$$w_i(x) = T \left(\frac{\Delta_i(x)}{\Delta_{(q)}(x)} \right),$$

where $\Delta_i(x)$ is the distance from x_i to x . For $\alpha > 1$, the $w_i(x)$ are defined in the same manner, but $\Delta_{(q)}(x)$ is replaced by $\Delta_{(n)}(x)\sqrt{\alpha}$.

After the neighborhood weights are computed, the local fitting is carried out. If locally linear fitting is used, a linear surface is fitted to the data using weighted least squares. This means that we find values of a , b , and c that minimize

$$\sum_{i=1}^n w_i(x)(y_i - a - bu_i - cv_i)^2.$$

Let $\hat{a}$, $\hat{b}$, and $\hat{c}$ be the values that achieve the minimization. Then the loess fit at $x = (u, v)$ is

$$\hat{a} + \hat{b}u + \hat{c}v.$$

If locally quadratic fitting is used, a quadratic surface is fitted to the data in a similar manner using weighted least-squares. In other words, the terms 1 , u , v , uv , u^2 , and v^2 are fitted locally.

If bisquare is added to loess fitting, an initial loess estimate is computed using the above procedure. Then bisquare iterations are carried out using bisquare weights, r_i , until the surface converges. For the local fit at x , the weight for (x_i, y_i) is $r_i w_i(x)$, the bisquare weight times the neighborhood weight.

If a loess fit, $\hat{g}(u, v)$, has been specified to be conditionally linear in u , then given any value of v , say $v = \cdot$, $\hat{g}(u, \cdot)$ is a linear function of u . Similarly, if the fit is conditionally quadratic in u , $\hat{g}(u, \cdot)$ is a quadratic function of u . Suppose first that $\lambda = 1$. To make the fit conditionally linear in u , we simply ignore the measurements of u in computing the distances, $\Delta_i(u, v)$, from $x = (u, v)$ to $x_i = (u_i, v_i)$. In other words, only the measurements of v are used in the computation of distance. Given $v = \cdot$, the distances $\Delta_i(u, \cdot)$ are the same for all u , so the function fitted locally at $(u, \cdot)$, which is linear, remains the same as u changes, and thus this linear fit is equal to $\hat{g}(u, \cdot)$. Suppose now that $\lambda = 2$. The same method makes it conditionally quadratic in u — the u_i are ignored in computing the distances. For $\lambda = 2$, the fit can be made conditionally linear in u by dropping the term u^2 in the local fitting. In other words, the terms $1, u, v, uv$, and v^2 are fitted instead of $1, u, v, uv, u^2$, and v^2 . Again, of course, the u_i are ignored in computing distance.

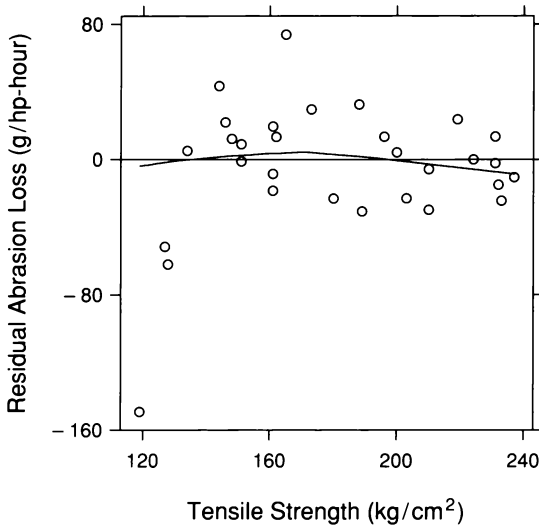
In later chapters, loess will be used to fit data with more than two factors. Except for one issue, the above discussion holds with obvious modifications. The exception is the definition of $w_i(x)$ for $\alpha > 1$. For p factors, $\Delta_{(q)}(x)$ is replaced by $\Delta_{(n)}(x) \sqrt[p]{\alpha}$.

4.4 Graphing Residuals

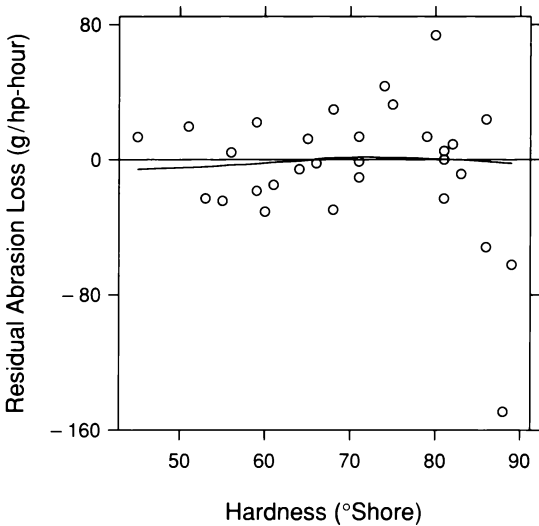
The overall strategy for graphing residuals for trivariate data is the same as that for bivariate data. First, residual dependence plots are made to check for lack or surplus of fit. If a problem is detected, we can either alter the fit or simply use the visualization of the residuals as part of the characterization of how the response depends on the factors. Other residual displays explore the distribution of the residuals. An s-l plot can detect monotone spread; if it occurs, we can attempt to remove it by a power transformation of the response. A normal q-q plot of the residuals compares the distribution of the residuals with the normal distribution. An r-f spread plot shows how much of the variation in the response is explained by the fit, and how much remains in the residuals.

Searching for Lack of Fit with Residual Dependence Plots

A good first step to probe lack of fit is to graph the residuals against each of the factors. This is done in Figures 4.18 and 4.19 for the fit to the rubber data. The three negative outlying residuals correspond to the three aberrant observations. Their abrasion losses are lower than expected based on the pattern of the other data, so their residuals are negative. But otherwise no lack of fit is revealed by these residual dependence plots.

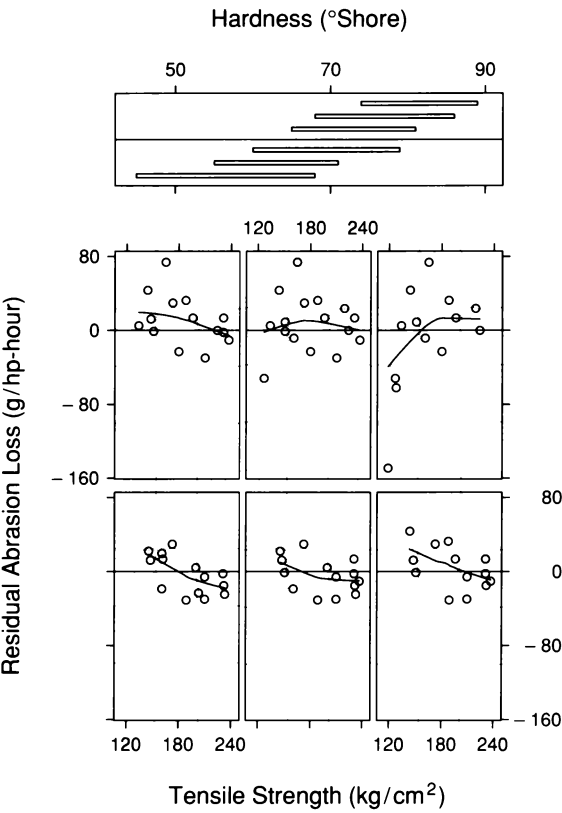


4.18 The residual dependence plot graphs the abrasion loss residuals against tensile strength. The parameters of the robust loess curve on the graph are $\alpha = 1$ and $\lambda = 1$.



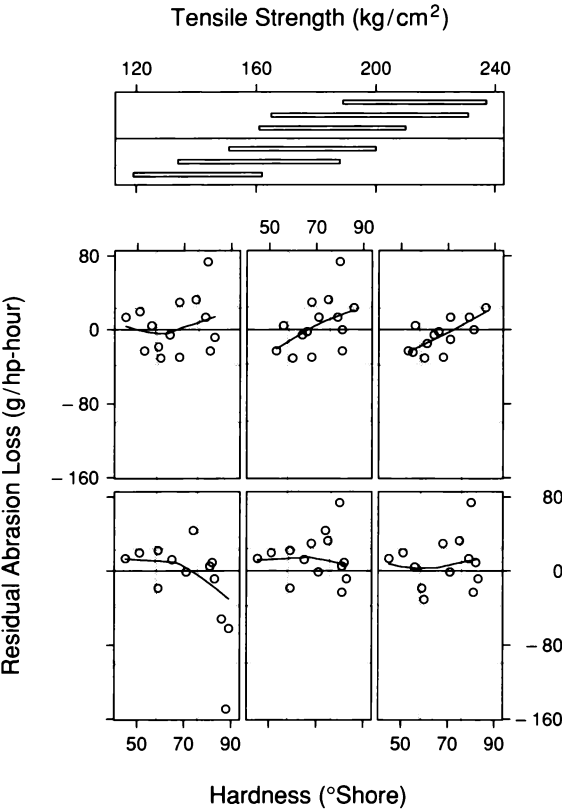
4.19 The residual dependence plot graphs the abrasion loss residuals against hardness. The parameters of the robust loess curve are $\alpha = 1$ and $\lambda = 1$.

Figure 4.20 is a coplot of the residuals against tensile strength given hardness. The intervals of hardness are the same as those used in the coplot of the data, Figure 4.3. The coplot reveals a remaining dependence in the residuals, which signals lack of fit. The underlying pattern on panels (1,1), (2,1), (3,1), and (1,2) has a negative slope that results largely from positive residuals at tensile strengths below 180 kg/cm², the breakpoint of the hockey-stick fit. This means the slope of the fit below the breakpoint is not sufficiently negative. Despite the bisquare fitting, the three outliers have caused lack of fit by forcing a reduction in the steepness of the fitted slope below the breakpoint. This is disappointing since bisquare is normally impervious to a small fraction of outliers. But in this example, the three outliers have conspired together at a location that exerts substantial leverage on the fit. It is a good lesson. No fitting method is perfect. This is one reason why we must subject residuals to intensive visualization.



4.20 The abrasion loss residuals are graphed against tensile strength given hardness. The parameters of the robust loess curves are $\alpha = 1$ and $\lambda = 1$. The aspect ratio of the dependence panels is 2 to accommodate the outlying negative residual.

Figure 4.21 is a coplot of the residuals against hardness given tensile strength. Again, the intervals are the same as those used in the coplot of the data, Figure 4.4. More lack of fit is revealed — an interaction between hardness and tensile strength. As tensile strength increases, the underlying pattern has an increase in slope. The effect is convincing, though small compared to the overall variation in abrasion loss, which is why we did not detect it on the original coplots of the data. This represents another example of the magnifying effect of residual plots, which was discussed in connection with the Playfair data in Chapter 3. There is another quite interesting aspect to the pattern of the interaction — the three aberrant observations appear to be conforming to it, but in a more extreme way.



4.21 The abrasion loss residuals are graphed against hardness given tensile strength. The parameters of the robust loess curves are $\alpha = 1$ and $\lambda = 1$. The aspect ratio of the dependence panels is 2 to accommodate the outlying negative residual.

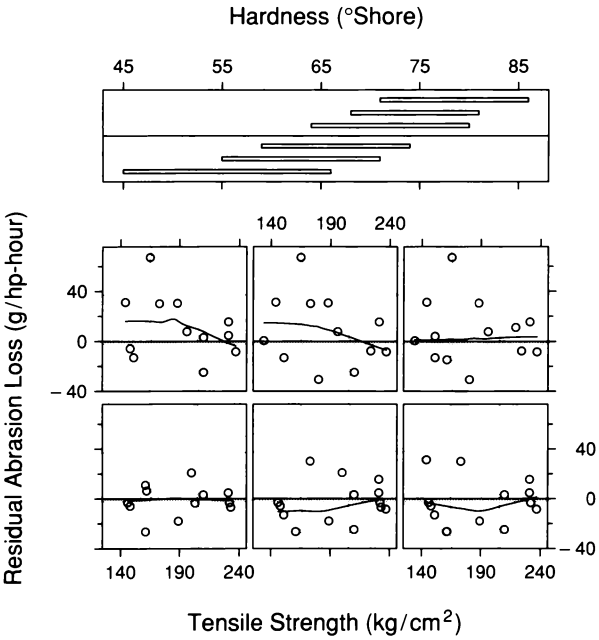
The visualization has given us substantial qualitative insight into the behavior of the rubber data. There is a hockey-stick pattern of dependence on tensile strength and a linear dependence on hardness. For most of the data, there appears to be a mild interaction between hardness and tensile strength in which the hardness slope increases somewhat as tensile strength increases. Three observations in one corner of the factor space show aberrant behavior that might represent a conforming to the interaction pattern. For some purposes, this qualitative description might be enough, for example, if the 30 runs that generated the rubber data were just a pilot experiment to plan a more thorough one.

Suppose, however, that we want a quantitative description of the dependence of abrasion loss on hardness and tensile strength. Then we must continue the fitting. It is unrealistic to suppose that we could account with much accuracy for the dependence in the region of the factor space occupied by the three aberrant observations. We will drop them. Since the residual coplot in Figure 4.21 showed an interaction, we must add an interaction term to the parametric family that we fit to the data. One possibility is the product term ht . But the residual coplot in Figure 4.20 showed the fit was adequate above the tensile strength breakpoint of 180 kg/cm^2 , and the product term would affect the fit at all values of tensile strength. Instead, we will add $h[t - 180]^-$, which has no effect above the breakpoint, but allows for an interaction below it. With this term, the effect of h given t is still linear, as the residual coplot in Figure 4.21 suggests it should be, but with a changing slope. The bisquare fit to the reduced data with the added term is

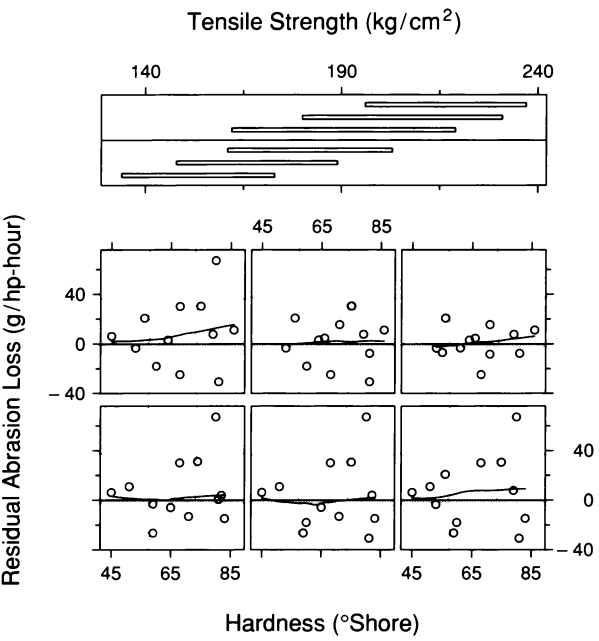
$$\hat{g}(h, t) = 531 - 5.78h - 7.76[t - 180]^- + 0.055h[t - 180]^- .$$

For this fit, the slope of $\hat{g}(h, t)$ as a function of h increases as t increases up to 180 kg/cm^2 , and then remains constant.

Figures 4.22 and 4.23 are coplots of the residuals for this new fit. No convincing dependence is revealed; the new fit appears adequate.

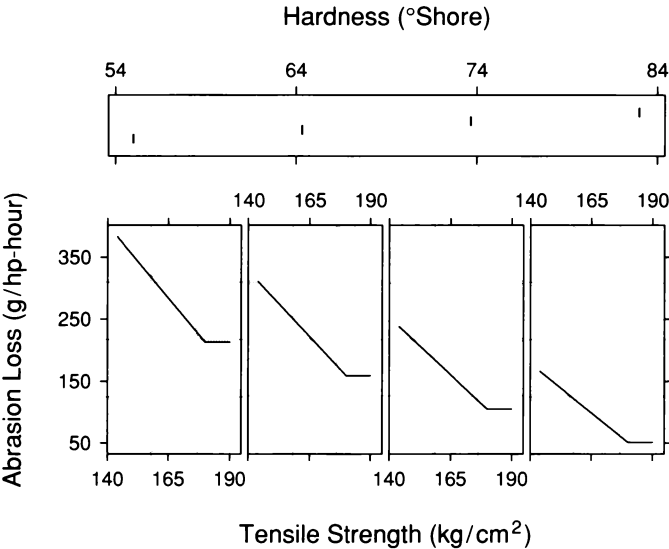


4.22 The residuals for the interaction fit to the reduced rubber data are graphed against tensile strength given hardness. The parameters of the robust loess curves are $\alpha = 1$ and $\lambda = 1$, and the target fraction of overlap of the intervals is 3/4.

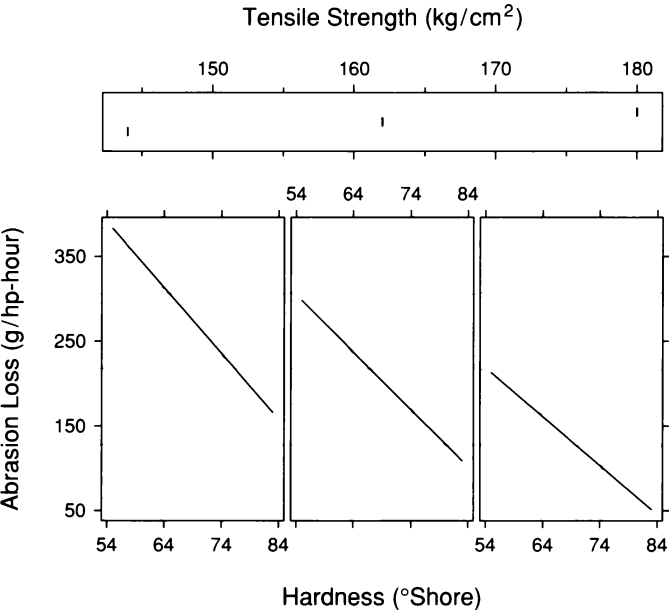


4.23 The residuals for the interaction fit to the reduced rubber data are graphed against hardness given tensile strength. The parameters of the robust loess curves are $\alpha = 1$ and $\lambda = 1$, and the target fraction of overlap of the intervals is 3/4.

Figures 4.24 and 4.25 are coplots of the new fit with the same conditioning values as the coplots of the first fit in Figures 4.16 and 4.17. The interaction is visible, although small, as expected.



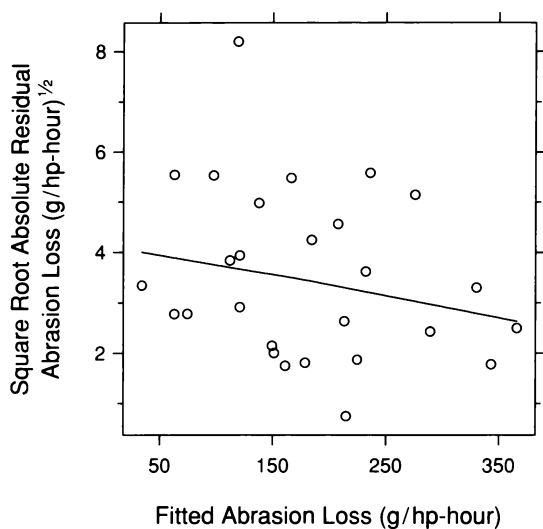
4.24 A coplot graphs the fitted abrasion loss surface against tensile strength given hardness. The segments of the broken lines below 180 kg/cm², the breakpoint, are banked to 45°.



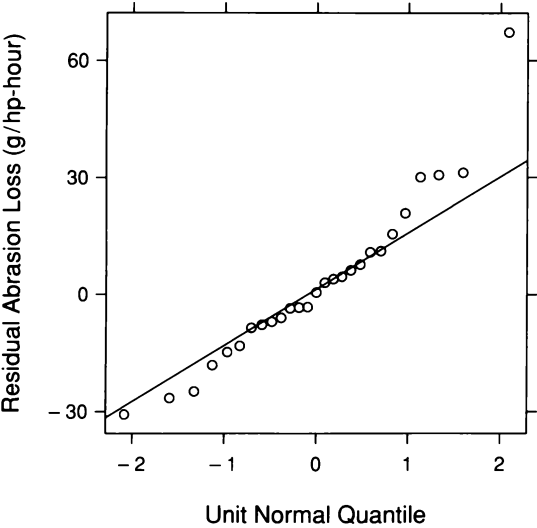
4.25 A coplot graphs the fitted abrasion loss surface against hardness given tensile strength. The lines are banked to 45°.

The Distribution of the Residuals

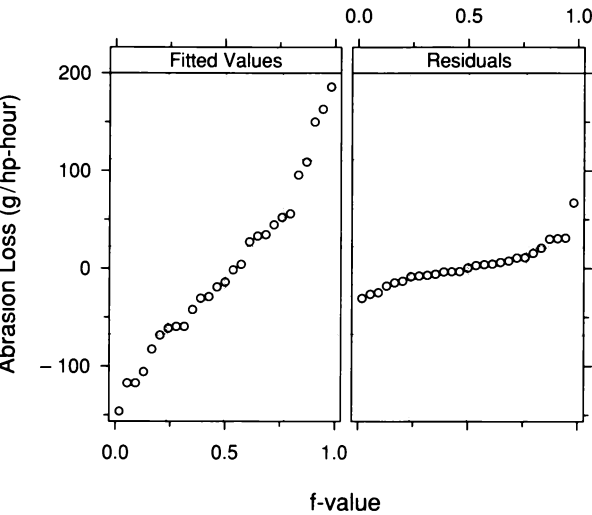
Figure 4.26 is an s-l plot for the fit to the reduced rubber data. The loess curve has a slight negative slope, but in view of the small number of observations, the magnitude of the effect is too small for us to believe there is significant monotone spread. Figure 4.27 is a normal quantile plot of the residuals; there is mild skewness toward large values. Figure 4.28 is an r-f spread plot. Despite the difficulties with the data, the response has been closely fitted for the reduced data.



4.26 An s-l plot checks for monotone spread in the abrasion loss residuals. The parameters of the robust loess curve on the plot are $\alpha = 2$ and $\lambda = 1$.



4.27 The distribution of the abrasion loss residuals is displayed by a normal q-q plot.

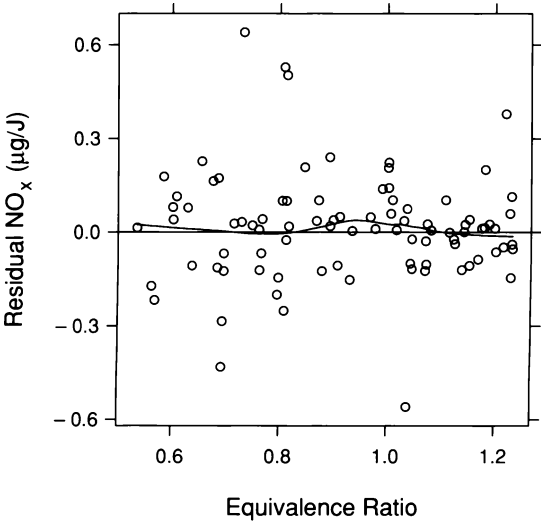


4.28 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the interaction fit to the reduced rubber data.

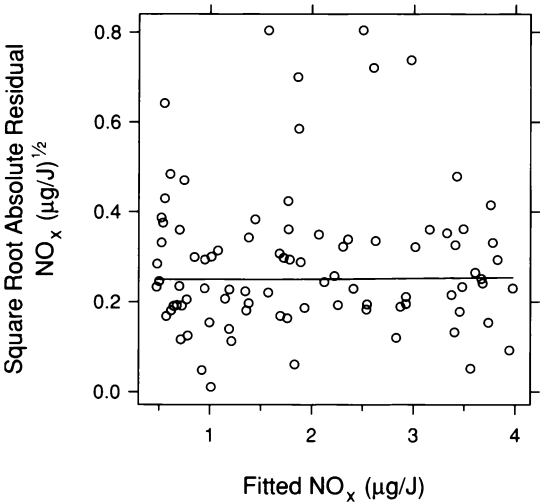
Ethanol Residuals

Fumblings every bit as extensive as those for the rubber data produced the ethanol fit graphed in the coplots of Figures 4.12 and 4.13. The tortuous path will not be described here. The robust fit, whose loess parameters are $\alpha = 1/3$ and $\lambda = 2$, appears to be adequate. Four residual dependence plots were used to check for lack of fit — two scatterplots of the residuals against the two factors, a coplot of the residuals against C given E , and a coplot of the residuals against E given C . No lack of fit was revealed. Increasing the value of α introduces lack of fit. This is illustrated in Figure 4.29. The residuals are graphed against E for a fit with α increased to $1/2$. There is a slight positive bump at $E = 0.9$, the approximate location of the ridge in the NO_x surface. Thus the fit is not reaching the top of the ridge. While the lack of fit is small, there is no reason to introduce it because, as the coplots in Figures 4.12 and 4.13 showed, the surface with $\alpha = 1/3$ is quite smooth.

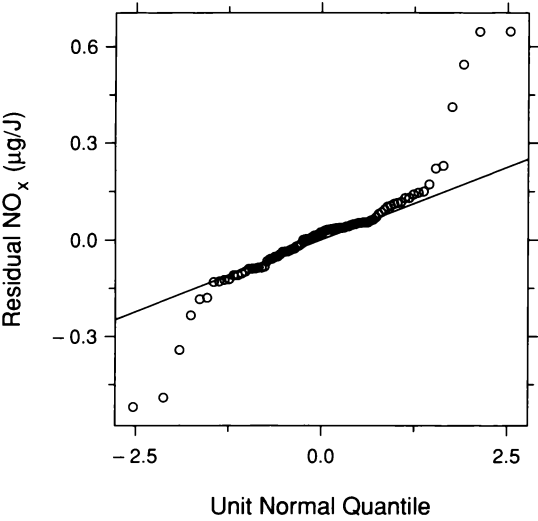
Figure 4.30 is an s-l plot. No monotone spread is revealed. Figure 4.31 is a normal quantile plot of the residuals. There is leptokurtosis, but in this case bisquare has done its job. Figure 4.32 is an r-f spread plot. The fit accounts for much of the variation in the NO_x concentrations.



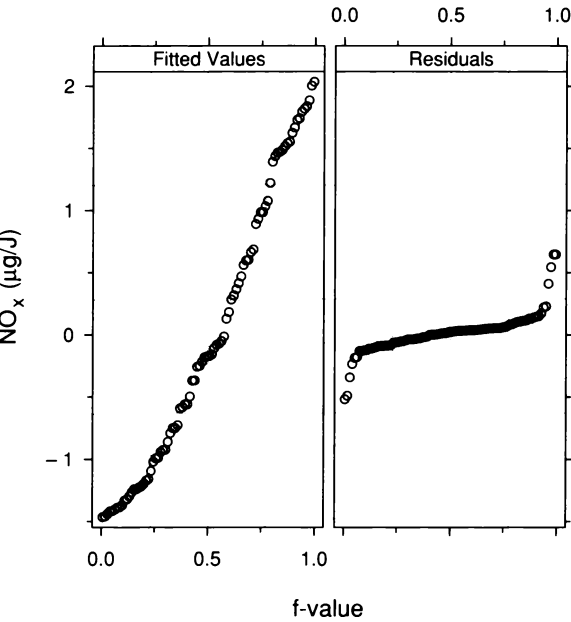
4.29 The residuals from a smoother robust loess fit to NO_x are graphed against equivalence ratio. The parameters of the robust loess curve on the graph are $\alpha = 1/2$ and $\lambda = 1$.



4.30 An s-l plot checks for monotone spread in the residuals for the robust loess fit to NO_x. The parameters of the robust loess curve on the plot are $\alpha = 2$ and $\lambda = 1$.

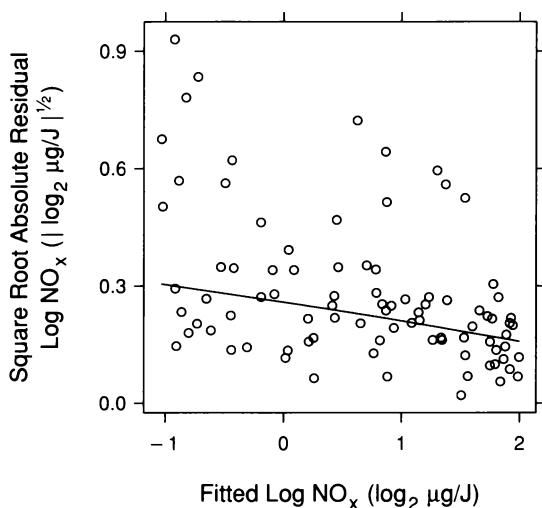


4.31 A normal q-q plot compares the normal distribution with the distribution of the residuals for the robust loess fit to NO_x .



4.32 An r-f spread plot compares the spread of the residuals and the fitted values minus their mean for the robust loess fit to NO_x .

In the original analysis of these data, the experimenter fitted log concentration by a high-order polynomial and used least-squares [12]. The leptokurtosis in the residuals makes it clear that least-squares is inappropriate. And Figure 4.33 shows that taking logs induces monotone spread; the figure is an s-l plot for a loess fit to log concentration using the same parameters as the fit to concentration, and using bisquare. The spread of the residuals decreases with increasing location, exactly what one would expect when logs are taken in a case where there is uniform spread for a fit to the data without transformation. The mistreatment of these data by the experimenter and others who analyzed them will be taken up at the end of this chapter.

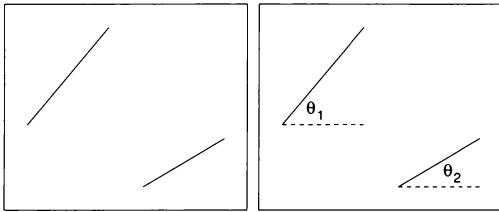


4.33 An s-l plot checks for monotone spread for a robust loess fit to log NO_x. The parameters of the robust loess curve on the plot are $\alpha = 2$ and $\lambda = 1$.

4.5 More on Coplots and Fitting

Judging the slopes of line segments on a graph is a critical visual task of graphical perception that yields information about the rate of change of one variable as a function of another [20]. The purpose of banking to 45° is to maximize the efficiency of such slope judgments. An experiment, whose data we will now analyze, led to the first formulation of the 45° principle [25, 26].

The left panel of Figure 4.34 has two line segments. Suppose we must judge the ratio of the slopes of the segments; more specifically, suppose we judge the ratio of the slope of the lower right segment to the slope of the upper left segment as a percent. Suppose the guess from a quick visual assessment is 40%. The true value is 50%, so our absolute error is $|40\% - 50\%| = 10\%$. What might determine the accuracy with which we make such a judgment?



4.34 The figure illustrates the judgment of slopes of line segments, an important task in the visual decoding of information from graphical displays.

Experiment

In the experiment, 16 subjects judged the relative slopes of 44 line-segment pairs. All slopes were positive, and the subjects judged what percent one slope was of another, just as we did for the segments on the left panel of Figure 4.34. The true percents ranged from 50% to 100%. Subjects judged this complete set 12 times, and the order of the line-segment pairs was randomized separately within each of the 12 replications. For each subject's judgment,

$$|\text{true percent} - \text{judged percent}|$$

was computed; then, for each line-segment pair, these absolute values were averaged across subjects and replications. The resulting 44 values of this response variable, a , measure the absolute error of judging the relative slopes of the line segments.

There are three factors in the experiment, all of which measure geometric aspects of the segments. To describe them, we need the notation in the right panel of Figure 4.34; θ_1 and θ_2 are the *orientations* of the two line segments. The first factor is p , the actual value of the percent being judged. The second factor is

$$d = |45^\circ - (\theta_1 + \theta_2)/2| ,$$

the distance of the mid-orientation, $(\theta_1 + \theta_2)/2$, of the two segments from 45° . Values of p and d determine the orientations of the segments; that is, if we specify both d and p , we specify the orientations of the segments, and therefore the ratio of the slopes. In the experiment, d and p were systematically varied. Another factor is the resolution, $r = \theta_1 - \theta_2$, of the two orientations, which is related to p and d by

$$r = \arcsin \left[\frac{100 - p}{100 + p} \cos(2d) \right] .$$

The important variable in the experiment is d . When $d = 0$, the segments are banked to 45° . And in making a graph, we control d by the aspect ratio. The hypothesis of the experiment was that for fixed p , the absolute error, a , increases as d increases; in other words, a increases as the segments get further from the position prescribed by 45° banking. As d increases for a fixed p , r decreases. The hypothesis is based on the notion that it is harder to judge a slope ratio p as the visual separation of the segments decreases.